

Time for Finite Systems

Coherence-Rupture-Regeneration as a Derived Clock for the Free Energy Principle

Implications, cross-disciplinary value, and a 2026–2030 financial assessment of the CRR temporal grammar — spanning oncology, the Levin research programme, human societies, machine intelligence, and a scale-invariant account of temporal relationality in finite systems.

Prepared for	The directorate of a frontier science laboratory with an active cross-disciplinary network
Subject corpus	A. Sabine, CRR working papers and syntheses (github.com/alexsabine/Ryan_CRR), incl. the AGI 2026 conference paper
Assessment basis	Close reading of the five-document corpus; live survey of 2026 industry and research bottlenecks (sources in Appendix B)
Reading offered	Dual: (i) CRR as an organising heuristic in its current form; (ii) CRR as a falsifiable empirical programme, 2026–2030
Status	Exploratory synthesis. CRR is unvalidated; all financial figures are illustrative, probability-weighted scenario ranges.

Executive Summary

The Free Energy Principle (FEP) is a physics of persistence: it describes how self-organising systems hold a non-equilibrium steady state, and it is chronically criticised for being too general to falsify. Coherence-Rupture-Regeneration (CRR) proposes the piece FEP conspicuously lacks — a **derived clock**: a principled statement of when a regime is used up, what falls at that instant (exactly one bit: *it happened*), and how the next regime is rebuilt from a salience-weighted past. One dimensionless dial, $\Omega = \varepsilon/\ell$, is claimed to govern capacity, memory breadth, and rupture timing simultaneously — with the denominator derived from symmetry class (never fitted) and the numerator measurable from the system's own rupture-interval statistics.

This dossier assesses what that clock is worth. Three findings organise everything that follows.

First, the clock is where falsifiability enters the FEP programme. FEP alone makes no point predictions about succession between regimes; FEP + CRR predicts Gamma-distributed rupture intervals ($CV = \sqrt{\Omega}$) and demands that a single Ω , read three independent ways — from the consolidation trigger, from the memory kernel, and from interval statistics — agree. That three-way consistency requirement is a genuine strong-inference test, cheap to run, with pre-stated falsifiers.

Second, industry and science in 2026 are independently converging on CRR-shaped problems without a theory. AI agent-memory vendors ship 'sleep-time' consolidation on improvised schedules; frontier labs measure 15–32% capability loss under continual fine-tuning; adaptive cancer therapy schedules treatment holidays at arbitrary 50%/100% burden thresholds; the Levin programme documents bistable somatic pattern memories with no interval law; thermodynamic-chip makers state plainly that their hardware is ahead of its workloads; and seismic-hazard agencies already price earthquake recurrence with an accumulation-to-threshold interval law whose governing parameter is the CV of the rhythm — a CRR-shaped clock in production use. Every one of these is a scheduling-and-salience gap — precisely the gap CRR claims to fill.

Third, the value is real but conditional, and the option is cheap. Our probability-weighted scenario analysis puts plausible value capture for a well-networked frontier lab at roughly **US\$27–155M cumulative over 2026–2030** (central estimate $\approx$ US\$60M), dominated by AI-side applications where falsification takes months, with a post-2030 option tail in oncology, bioelectronics, and thermodynamic computing plausibly an order of magnitude larger. The entire empirical ladder through 2027 runs on the laboratory's own research team with AI-assisted execution, so marginal cost is dominated by staff time and compute. The asymmetry — small stake, fast kill-criteria, large multi-domain upside, and a lingua-franca instrument for a networked lab even in partial-success scenarios — is the investment case.

Recommended posture: fund the ladder, not the vision. Run the synthetic-arena signature test and the two AI-side trials (consolidation trigger; salience-weighted replay) in the next nine months; open two partner conversations (Levin laboratory; an integrated mathematical-oncology group) in parallel; treat the societal reading as heuristic language, valuable but not yet science. Section 7 gives the twelve-month plan.

1 The Grammar in One Page

CRR claims that anything finite that persists — a habit, a healing wound, a learning machine, a tissue, an institution — cycles through three operations. **Coherence**: it accumulates structure that works, counted as Fisher-Rao arc length in units of the smallest step it can resolve, $C = (1/\varepsilon) \int L \, d\tau$. Because Fisher-Rao distance is measured in units of distinguishability (for a Gaussian family, a z-score), C is a pure number — which is what makes the grammar unit-free and hence a candidate for scale-invariant application. **Rupture**: no finite system accumulates forever; when accumulated coherence exhausts the count its arena affords ($C \cdot \Omega = 1$), a cut falls — an instantaneous, contentless event carrying exactly one bit. **Regeneration**: the system rebuilds from its own past under a maximum-entropy salience kernel $e^{C/\Omega}$, which weights past moments by how coherent they were, not how recent — a salience kernel, not a decay kernel, and the source of the grammar's process-level non-Markovianity.

The single dial $\Omega = \varepsilon/l$ divides never-fitted from measured. The denominator l — the geodesic extent of the arena — is fixed by the symmetry class of the system's statistical manifold via Čencov's theorem (π for bistable Z_2 systems; $2\pi \cdot \sqrt{I}$ for rotational $SO(2)$ systems, a parameter-free ratio of 2). The numerator $\varepsilon = 1/\sqrt{N}$ is the system's resolution by Cramér-Rao, and it is *measurable*: the quantal clock implies rupture intervals are Gamma-distributed with shape $1/\Omega$, so Ω equals the squared coefficient of variation of the system's own rupture intervals. Low Ω : a few crystallised memories dominate every rebuild — habit, expertise, rock. High Ω : the whole past votes — play, exploration, water. Everything that learns sits strictly between, and development is annealing: ε falls as evidence per distinction grows.

Three further results matter for this dossier. (i) Each rupture costs at least $k_B T \ln 2$ by Landauer — the bit acquired at the edge — making any finite accumulator a *mortal computer* by construction. (ii) Mortality is nested: consolidation is a small death with a small inheritance, and the same kernel that schedules sleep decides what sleep replays; hardware death is the limiting case where the student is another body. (iii) Coupling is quantitative: a partner absorbing fraction f of a learner's traverse raises its effective openness to $\Omega_{\text{eff}} = \Omega/\sqrt{1-f}$ — 'holding' as an engineering specification, with graduated withdrawal as a schedule and two named failure modes.

Provenance note: the corpus comprises one peer-reviewed conference paper (AGI 2026, reporting phase-gating between precision channels, $\chi^2 = 8,041$) and four exploratory working papers/syntheses, two of which disclose AI-assisted drafting under the author's direction. The corpus itself flags an unresolved convention (arc-length vs. Fisher-action clocks) and states falsifiers for every prediction. This dossier treats that candour as an asset and holds the programme to it.

2 What a Derived Clock Gives the Free Energy Principle

FEP describes systems that hold together. Its formal object, the non-equilibrium steady state, is a description in which — on average — nothing changes: entropy is produced, flows persist, the regime holds. The framework is therefore superb on persistence *within* a regime and silent on succession *between* regimes. Four consequences follow if a derived clock of CRR's type holds up.

2.1 A unit of analysis for change

Regimes become countable, individuated objects with beginnings, endings, durations, and inheritance relations. FEP acquires a native account of development, consolidation, death, and succession — a physics of *biography* layered on its physics of persistence — without any modification to the mechanics. The division of labour is clean: variational free energy says what the system updates toward; the clock says when a way of being is complete and what its completion bequeaths. This is the difference between describing a heartbeat and describing a life.

2.2 Falsifiability enters through the clock

The chronic complaint against FEP — that it is a principle, not a theory, and can absorb any observation — is answered not by defending FEP but by composing it. FEP + CRR makes point predictions FEP alone cannot: rupture intervals Gamma-distributed with shape $1/\Omega$, $CV = \sqrt{\Omega}$, skewness $2\sqrt{\Omega}$; a CV ratio of exactly $\sqrt{2}$ between matched bistable and rotational systems; and, crucially, the **three-readings test** — the Ω inferred from interval statistics must equal the Ω governing the consolidation trigger and the Ω governing the memory kernel. A system in which the three disagree has falsified the grammar's central claim that capacity, breadth, and timing are one property. Composite falsifiability of this kind is exactly what the active-inference ecosystem has lacked, and it is testable within weeks in synthetic arenas.

2.3 Licensed non-Markovian memory

The regeneration kernel weights the past by lived coherence, not recency, making the process history-dependent at the event level while leaving the underlying mechanics Markovian. This gives FEP a principled bridge to phenomena it currently handles awkwardly — trauma (a low- Ω regime's closing cells dominating every reconstruction), expertise, institutional memory, and the empirical finding in the AGI 2026 study that coherence-weighting is essentially uncorrelated with reinforcement-learning temporal discounting ($r = -0.182$) yet assigned credit twice as accurately at depth 50 with $\sim 17\times$ fewer updates.

2.4 Scale composition, and the cost of the cut

Because completed events at one scale become historical material at another (the corpus's Axiom 5), nested Markov blankets acquire nested clocks that compose without a global timekeeper — development, sleep, and death become the same operation at three scales. And because each cut carries a Landauer floor, the clock prices itself in the currency stochastic thermodynamics already spends: C is not a metaphorical odometer but thermodynamic length, and $C \cdot \Omega = 1$ reads as a budget statement. The principal risks are stated in Section 6: the unresolved clock convention, the unestablished symmetry class of wild high-dimensional manifolds, and the nontrivial cost of online arc-length estimation at scale.

3 Domain Analyses

3.1 Cancer research

The fit, in one sentence: oncology already contains a clock discipline without a clock theory, and a boundary discipline without a timing variable.

(a) Adaptive therapy is an applied scheduling problem. The adaptive-therapy programme (Gatenby, Anderson, Zhang and colleagues; Moffitt's Integrated Mathematical Oncology group) schedules treatment holidays so that drug-sensitive clones competitively suppress resistant ones; phase-2 results in metastatic castrate-resistant prostate cancer extended time-to-progression while substantially reducing cumulative drug use. Yet the field's own survey literature concedes the standard protocol — treat to 50% burden, pause until 100% — uses arbitrary thresholds that ignore patient-specific tumour dynamics, the acknowledged source of outcome heterogeneity; recent work explicitly asks how to derive optimal treatment *timing* by matching models to per-patient dynamics. CRR's offer is concrete: read each patient's Ω from the coefficient of variation of their burden-oscillation intervals (PSA time series from completed trials already exist), and place switch points at exhaustion on the tumour-treatment system's own geometry rather than at fixed percentages. This is testable *retrospectively* in 12–24 months by the existing research team.

(b) Dormancy and relapse as an interval law. If tumour regimes obey the quantal clock, relapse-interval distributions across patient cohorts should be Gamma-family with cohort-readable shape $1/\Omega$ — a falsifiable statistical signature sitting in registry data, and a candidate timing prior for surveillance scheduling. A clean negative here would be genuinely informative about the grammar's biological reach; the corpus's own *E. coli* episode (an early interval law failed cleanly and was replaced by the quantal law, which re-read the same data as a measurement) sets the precedent and the discipline.

(c) The Levin-flavoured reading: cancer as a nesting failure. On the bioelectric account, cancer cells shrink their computational boundary and disconnect from the tissue-level network, reverting toward unicellular goal-space. In CRR terms this is a clock secession: a subsystem stops inheriting the collective's regime, drops out of the coupled oscillation, and runs its private, faster cycle — the formal shadow of gap-junction decoupling. The reframe matters because it points interventions at *re-entrainment* (restore coupling; raise the collective's share f of the cell's traverse) rather than exclusively at kill-rate — which is precisely the morphochemical bet, now with a scheduling variable attached. Status: hypothesis-generating; the bridge-mapping problem (what is the tumour's manifold? what is its ϵ ?) is real and stated.

3.2 The Levin research programme

This is the deepest structural affinity in the dossier, and the door is already ajar: the CRR bibliography cites Watson-Levin on the collective intelligence of evolution and development, and the Tissot-Levin-Buckley-Watson minimal-agent result in which *timing rather than magnitude* is the operative variable is the corpus's own template for coupled clocks.

Point-for-point: Levin's multi-scale competency architecture *is* nested regimes with inheritance (Axiom 5). Morphogenesis as navigation of anatomical morphospace *is* a trajectory on a statistical

manifold accumulating Fisher-Rao arc length. TAME's cognitive light cone — the spatial reach of a system's goals — acquires a temporal twin with a number attached: Ω as the depth of accessible history and breadth of accessible futures (the AGI 2026 paper makes this identification explicitly). Regeneration is literally the grammar's third verb, and planarian regeneration supplies the killer experiment: the Levin group has documented **bistability of somatic pattern memories** (stochastic one-head/two-head outcomes in bioelectrically perturbed planaria) — a Z_2 symmetry class with a parameter-free π threshold. CRR predicts the interval statistics of stochastic transitions between pattern states ($CV = \sqrt{\Omega}$ at the tissue's resolution) and demands the Ω read there match the Ω governing the tissue's regeneration kernel. Xenobot and anthrobot developmental protocols, meanwhile, get the scaffolding law $\Omega_{\text{eff}} = \Omega/\sqrt{1-f}$: graduated withdrawal of experimenter constraint as an equation with named failure modes, rather than an art.

Strategic read: the Levin programme has extraordinary breadth of phenomena and a famous appetite for formal tools imported from other disciplines; what it does not have is a temporal instrument — a dial readable from a preparation's own rhythm that transfers between planaria, organoids, and synthetic constructs. If CRR's interval law survives one planarian dataset, the lab holding it owns a shared measurement standard across the entire basal-cognition network.

3.3 Human societies and organisations

Here the grammar is at its strongest as a heuristic and its weakest as science, and the two readings must be kept apart with discipline.

As heuristic, the yield is immediate. Institutions are finite systems: they accumulate procedure, exhaust a way of operating, and rebuild from selectively weighted institutional memory — and the grammar names design levers current practice conflates. Ritual and rites of passage read as *engineered small ruptures*: scheduled consolidation with curated reconstruction material ϕ , which is why societies that abolish transition rituals do not abolish transitions, only their scheduling. Kuhnian paradigm shifts read as regime exhaustion (normal science as coherence accumulation; anomaly crisis at $C \cdot \Omega \rightarrow 1$; reconstruction weighting the tradition's most coherent stretches). Polarisation and echo chambers get a formal description the corpus states outright: a mirror that reflects with high fidelity while absorbing none of the traverse *flattens nobody's kernel and can only sharpen what is already peaked* — Ω -collapse as the mechanism of radicalisation, with an implied design remedy (interlocutors must absorb traverse, i.e., $f > 0$, to raise anyone's effective openness). Legitimacy crises read as institutional still-face: well-formed outputs at non-participating timing, with the still-face meta-analytic literature describing what that does to the partner. And a 'shared present' becomes a coupled-oscillator fact rather than a metaphor — clocks lock when rate mismatch falls below twice the coupling, predicting a *cliff*, not a slope, in social coordination.

As empirical science, the honest rating is 'not yet.' Interval statistics of institutional regime transitions could in principle be confronted with cliodynamic and organisational datasets (constitutional lifetimes, firm reorganisation cadences, paradigm-turnover intervals), and the Gamma/CV signature gives a concrete thing to look for. But measurement of a society's ε is unsolved, confounds are brutal, and the just-so-story risk is maximal precisely where the narrative appeal is highest. The dossier's recommendation: deploy the vocabulary, publish the

essays, run one exploratory dataset study — and attach no financial weight beyond that.

3.4 Machine intelligence and computing (summary of the industrial survey)

A 2026 bottleneck survey conducted for this dossier found the same shape in seven areas: industry has built the *machinery* and improvises the *schedule and salience policy*. (i) Agent-memory vendors and frontier labs now ship background consolidation — 'sleep-time compute', Mem0's Dream, Anthropic's hippocampally-styled Dreaming — while stating openly that consolidation is not neutral and needs a policy for what counts as durable signal; triggers remain idle-time and periodic. (ii) Continual fine-tuning of frontier models degrades 15–32% of capability across task sequences, with 2025-era practice recommending 25–50% replay; the trigger (P1) and kernel (P2) are directly testable against matched budgets. (iii) Thermodynamic-computing leaders state their hardware is ahead of its workloads; CRR's regeneration integral is natively a TSU workload (set $E = -C$, $T = \Omega$; the archive becomes a programmable energy-based model — 'sampling one's own past at temperature Ω '). (iv) Analog/neuromorphic substrates suffer conductance drift the field treats as an enemy; the grammar re-reads drift as substrate speed, converting silent corruption into scheduled turnover. (v) Robot fleet learning names closing the deploy→learn loop as its central bottleneck, and the industry's supervisor-ratio glide path ($\approx 1:5 \rightarrow 1:50$) is graduated withdrawal awaiting P4's functional form. (vi) Voice-AI has discovered empirically that users report timing, not fidelity ('feels off', 'keeps pausing'); P5 predicts entrainment to the partner's rhythm beats fidelity improvements, with a cliff at lock-loss. (vii) Model collapse — recursive training losing distribution tails — is the echo-chamber theorem at industrial scale. Section 5 prices these.

3.5 The scale-invariant instrument: temporal relationality

The unifying asset is not any single application. Because coherence is arc length counted in units of the system's own resolution — z-scores, a pure number — the grammar is unit-free and therefore portable across twenty-two orders of magnitude of relaxation time (the corpus's Deborah-number calibration spans water to granite). 'When' is measured in a system's own cells rather than in seconds. That is what makes **temporal relationality** well-posed for the first time: two finite systems share a present exactly when their clocks lock; a caregiver, teacher, therapist, or institution 'holds' another system by absorbing a measurable fraction of its traverse; a completed regime at one scale is inheritance at the next. For a frontier lab whose value lies partly in its *network*, this is the pitch: a candidate lingua-franca measurement standard — one dial, three consistent readings, pre-stated falsifiers — that either dies quickly and cheaply, or becomes shared infrastructure across every collaboration the lab maintains, from oncology to organisational science. Instruments, not theories, are what disciplines adopt from one another; CRR is best understood, and best sold, as an instrument.

3.6 Financial modelling and markets

Quantitative finance already runs on regime language — Hamilton-style Markov-switching models are standard — but the transition hazards in those models are typically memoryless or hand-fitted, and the industry's deepest timing questions are answered by calendar. CRR supplies three candidates. **(a) A duration law.** If market regimes are coherence-accumulating, their lifetimes should be Gamma-distributed with shape $1/\Omega$ rather than exponential — directly testable

against volatility-regime and drawdown-interval datasets, with Ω read from the CV and nothing fitted. **(b) A fragility class.** The poised edge $s^* = 1 - \Omega$ — a regime maximally determined from inside with exactly one undetermined bit outside — is formal kin to the critical-slowness-down early-warning literature, and suggests indicator families for pre-transition states. **(c) The desk's own clock problem.** Alpha decay is regime exhaustion by another name, and 'when do we retire or refit this model?' is P1 transplanted: consolidate the strategy at exhaustion on its own information geometry, benchmarked against calendar and drawdown triggers at matched refit budget. The honest caveat is reflexivity: markets are adversarial, so any public clock signal gets arbitrated; the durable value is *internal* — model-lifecycle governance and scenario libraries weighted by coherence rather than recency — not signal sale.

3.7 Insurance and catastrophe risk

Insurance is the industry that prices interval distributions for a living, and it is large and liquid: the outstanding catastrophe-bond market reached a record $\approx$ US\$65.6B at mid-2026 on $\approx$ US\$18B of first-half issuance, inside an insurance-linked-securities market approaching US\$141B of limit. More striking for CRR: a grammar-shaped law is *already in production use*. Seismic-hazard practice long ago abandoned the memoryless Poisson model for renewal models — Brownian Passage Time — parameterised by mean recurrence and an 'aperiodicity' α that is literally the coefficient of variation of the interval rhythm (CV = 1 recovers Poisson; CV < 1 is quasi-periodic accumulation), with a generic worldwide $\alpha \approx 0.5$, Japan's agency value ≈ 0.24 , and fault studies spanning ≈ 0.3 – 0.7 . In CRR's units, $\alpha = \sqrt{\Omega}$: faults run at $\Omega \approx 0.06$ – 0.25 , the dial read off the planet's own rhythm. CRR's offers: an **Ω -audit of perils** — classify risks by interval CV into accumulation perils (earthquake, wildfire fuel load, infrastructure fatigue: CV < 1), memoryless shocks (CV ≈ 1), and over-dispersed clustered risks (CV > 1, where the grammar says no representable regime holds — a suggestive reading of contagion-type exposures); a **discriminating recurrence test** — CRR's quantal Gamma law versus BPT's diffusive inverse-Gaussian, near-identical at matched CV yet statistically separable, so the resonance becomes an experiment; and a **cycle clock** — the underwriting cycle itself read as institutional CRR, hard-market turns as ruptures and capacity rebuilt under a salience kernel of memorable losses. Data are abundant, retrospective, and cheap.

3.8 Two further applied cases

(a) Predictive maintenance and critical infrastructure. Physical assets are accumulation systems — fatigue is coherence — and maintenance is an engineered small rupture with an inheritance question (what carries across the overhaul). The predictive-maintenance market sits at $\approx$ US\$13–17B (2025–26) heading to roughly US\$30–56B by 2030 against an unplanned-downtime problem exceeding US\$1T annually, yet scheduling remains largely calendar- and threshold-based. The transplant is direct: exhaustion-triggered maintenance on the asset's own drift geometry, a fleet-level dial from failure-interval CV (separating fatigue-type from shock-type failure modes), and kernel-weighted decisions about what state survives an overhaul. **(b) Human performance and training periodisation.** Training stress accumulates; the deload is a scheduled rupture; supercompensation rebuilds weighted by the hardest quality sessions. Deload timing at exhaustion on the athlete's own accumulated arc — with readiness-rhythm CV (e.g., HRV) as the dial — is testable on existing wearable datasets for almost nothing, and doubles as the programme's most publicly communicable demonstration.

4 Two Readings: Heuristic Now, Empirical Programme 2026-2030

As a heuristic in its current form. CRR already functions as a disciplined checklist that practitioners across the surveyed fields have partially reinvented: What is this specific finite system's Ω — its dial, read from the CV of its own rhythm rather than assumed? That single question forces two more: its ε , the smallest distinction the system can resolve, and its l — the limit of the system, the finite extent that its arena and symmetry class afford. Does your consolidation trigger live on the system's own geometry or on an imposed calendar? Does your memory policy weight by salience or by recency, and what is its temperature? Who is absorbing traverse for whom, and on what withdrawal schedule? Is your coordination failure a content problem or a timing problem? This costs nothing, risks little beyond vocabulary, and is immediately deployable in the lab's advisory and design work. Its main hazard is premature reification — using the words while skipping the measurements — which the empirical ladder below is designed to prevent.

As an empirical programme. Everything hangs on the three-readings consistency test, approached by a speed-ordered ladder. Early rungs are decisive and fast; later rungs are partner-gated and slower. Run in-house with AI-assisted execution, the early rungs' marginal cost is staff time and compute rather than external spend.

#	Test (corpus ref.)	Domain	Duration	What it kills or confirms
1	Synthetic arenas: Z_2 vs $SO(2)$ learners with known l , ε ; Gamma interval law, $CV = \sqrt{\Omega}$, $\sqrt{2}$ CV ratio; three-readings consistency (P3)	Simulation	4-10 wks	The signature itself. Symmetric/left-skewed intervals, a CV ratio of 2, or Ω -disagreement kills the grammar at root.
2	Exhaustion-triggered consolidation vs matched-budget periodic, validation-triggered, and oracle schedules in task-free class-incremental learning (P1)	AI / continual learning	3-9 mo	No advantage over matched-budget periodic scheduling falsifies the trigger.
3	Replay sampled $\propto e^{C/\Omega}$ vs recency-, uniform-, and loss-weighted selection at equal buffer/budget (P2); same design inside an agent-memory stack	AI / agent memory	3-9 mo	No advantage over recency weighting at matched budget falsifies the kernel.
4	Timing entrainment vs output-fidelity manipulation in human-machine interaction; lock/unlock cliff (P5)	Voice / HMI	3-6 mo	Coordination insensitive to timing, or a smooth (non-cliff) transition, falsifies the coupling claim.
5	Regeneration as native workload: $E = -C$, $T = \Omega$ on thermodynamic-sampler simulation (thrml), then partner hardware	Thermo. computing	6-18 mo	Confirms/denies the 'kernel occupied, not computed' efficiency claim on real fluctuations.
6	Planarian pattern-memory bistability: interval statistics of stochastic state transitions; $\Omega(\text{intervals}) = \Omega(\text{kernel})$ (three-readings, in vivo)	Levin programme	12-24 mo	First biological confirmation — or a clean biological boundary for the grammar.
7	Retrospective adaptive-therapy analysis: patient Ω from burden-oscillation CV as predictor of time-to-progression; exhaustion-timed switch points in silico	Oncology	12-24 mo	Whether the dial adds predictive value beyond existing dynamical biomarkers.
8	Scaffolding law: teacher absorbing measured f of traverse; period invariance, breadth $\propto 1/\sqrt{1-f}$, graduated vs abrupt withdrawal (P4)	Robotics / curricula	9-18 mo	Falsified by f -dependent period, f -insensitive breadth, or unordered withdrawal schedules.

#	Test (corpus ref.)	Domain	Duration	What it kills or confirms
9	Exploratory: interval statistics of institutional regime transitions in cliodynamic / organisational datasets	Societies	flexible	Hypothesis-generating only; no financial weight attached.
10	Interval-law retrospectives: financial regime durations; insured-loss and earthquake recurrence (quantal Gamma vs diffusive BPT discrimination); machine-failure records — Ω from CV, nothing fitted	Finance / insurance / PdM	3–9 mo	Memoryless (exponential) durations, or CVs unrelated to trigger/kernel behaviour, bound the grammar's reach in these domains.

Table 1 — The empirical ladder, ordered by speed and decisiveness. Rungs 1–3 resolve inside twelve months and govern whether anything beyond heuristic use is warranted.

5 Financial Value Assessment, 2026–2030

5.1 Methodology and definitions

Figures are **illustrative, order-of-magnitude scenario estimates**, not forecasts. For each domain we estimate: (i) the **value pool** — the annual economic value of the scheduling/salience layer CRR addresses in that domain by 2030, anchored to third-party market data where it exists (Appendix B); (ii) **probability-weighted capture 2026–2030** — cumulative value plausibly accruing to a single well-networked frontier lab that runs the ladder and publishes/licenses results, combining grants, partnership revenue, method licensing, and attributable product value, weighted by our subjective probability that the relevant rungs succeed; (iii) **unweighted upside** — the success-scenario value without probability discount; and (iv) a **feasibility rating** (A–D, defined in Appendix A) for decisive validation *within the window*. Ranges are deliberately wide; treat ratios between rows, not absolute levels, as the information. Nothing here is investment advice.

5.2 Master table

Domain & CRR offer	Value pool addressed (by 2030)	Weighted capture 2026–30	Unweighted upside	Feas	First signal
1. AI agent memory: exhaustion trigger + salience kernel for background consolidation	Orchestration & state-mgmt layer $\approx$ \$1.1B (2025) growing inside a \$24–52B agentic-AI market; consolidation sub-layer est. \$2–6B/yr	\$8–40M	\$150M+ if the trigger/kernel become a de-facto standard	A	3–9 mo
2. Foundation-model continual learning: when to consolidate; what to replay (P1/P2)	Avoided-retraining and replay-efficiency pool est. \$0.3–1B/yr by 2029–30 (frontier runs 10^8 – 10^9 ; 15–32% forgetting; 25–50% replay budgets)	\$5–30M	\$100M+ via adoption in frontier post-training pipelines	A–	6–12 mo
3. Thermodynamic computing: regeneration as native TSU workload ($E = -C$, $T = \Omega$)	Field capitalisation 2026–30 est. \$0.3–1B (incl. \$75M US-CHIPS LOI); algorithm layer 10–20% of value; workloads are the stated bottleneck	\$2–15M	\$50–200M option if TSUs scale and CRR becomes a canonical workload	B	6–18 mo

Domain & CRR offer	Value pool addressed (by 2030)	Weighted capture 2026-30	Unweighted upside	Feas	First signal
4. Oncology: Ω -from-CV scheduling biomarker; exhaustion-timed adaptive-therapy switching; relapse-interval priors	Per-indication drug-cost savings + PFS value est. \$0.5–2B/yr across indications (adaptive dosing already halves cumulative drug use in prostate Ph-2); largely post-2030	\$3–12M	\$100M+ post-2030 licensing a validated timing biomarker (outside window)	C+	12–24 mo, retro.
5. Levin programme: interval law for pattern-memory bistability; Ω as shared dial; scaffolding law for synthetic morphology	Basal-cognition/bioelectrics programme funding est. \$50–150M over window (centre-scale grants, philanthropy); morphochemical option beyond	\$2–10M	\$30–80M if Ω becomes a standard measurable across the network	B	12–24 mo
6. Human-machine timing: rhythm entrainment > fidelity; lock-cliff design rule (P5)	Perceived-quality gap gating a claimed ~\$80B/yr conversational-AI labour displacement; timing IP addressable slice est. \$50–300M/yr	\$1–8M	\$30M+ licensing entrainment design to voice platforms	B+	3–6 mo
7. Robotics: scaffolding law (graduated withdrawal) for fleet learning and teleop curricula (P4)	Supervisor-ratio opex ($\approx 1:5 \rightarrow 1:50$ glide path) across emerging humanoid/fleet deployments; est. \$0.1–0.5B/yr by 2030	\$1–6M	\$40M+ as curricula standard in fleet-learning stacks	B	9–18 mo
8. Societies & organisations: heuristic frameworks; ritual-as-scheduled-rupture; echo-chamber formalism	Transformation-advisory market >\$100B/yr but scientifically shallow; rigorously addressable slice small in window	\$0.5–4M	\$10–30M for a validated timing diagnostic for institutions	D	24+ mo
9. Financial modelling: regime-duration law; poised-edge fragility class; model-lifecycle trigger (alpha decay = exhaustion)	Model-risk, refit-efficiency and regime-analytics value at systematic funds; est. \$0.2–1B/yr addressable	\$2–12M	\$50M+ as a model-lifecycle governance standard at major quant firms	B+	4–9 mo
10. Insurance & cat risk: Ω -audit of perils by interval CV; Gamma-vs-BPT recurrence; underwriting-cycle clock	Pricing/reserving edge on accumulation perils inside a record ~\$65B cat-bond market (~\$141B ILS limit); seismic practice already uses CV-parameterised renewal laws	\$2–10M	\$40–100M if the CV-read dial becomes a rating/pricing input	B+	6–12 mo
11. Predictive maintenance: exhaustion-triggered servicing on asset drift; fleet dial from failure-interval CV; overhaul-inheritance kernel	Slice of a \$30–56B/yr PdM market by 2030, against a >\$1T/yr unplanned-downtime problem; scheduling today is calendar/threshold-based	\$1–8M	\$50M+ as a scheduling layer in asset-performance platforms	B+	6–12 mo

Table 2 — Master financial table. Probability-weighted capture is cumulative 2026–2030 to a single well-networked frontier laboratory; see §5.1 for definitions and Appendix A for the feasibility scale. All figures are illustrative scenario ranges in 2026 US dollars. Rows 9–11 cover financial markets, insurance, and predictive maintenance.

5.3 Portfolio totals and phasing

Summing the probability-weighted rows gives a cumulative 2026–2030 range of **$\approx$ US\$27.5M (all-conservative) to $\approx$ US\$155M (all-favourable)**, with a central estimate near **US\$60M**. The distribution is strongly right-skewed: the modal path is a respectable grants-and-methods business concentrated in rows 1–2, while the tail is carried by option-like positions (rows 3–5) whose window value understates their post-2030 worth by roughly an order of magnitude.

Correlations matter and mostly help: rung-1 success de-risks every subsequent row simultaneously, which is why the ladder is sequenced as it is.

Cumulative expected value (\$M)	2026	2027	2028	2029	2030
Conservative path	0.4	2.5	8.5	17	27.5
Central scenario	1	6	19	38	60
Favourable path	2.5	15	47	96	155

Table 3 — Illustrative cumulative phasing of probability-weighted value. 2026 is essentially ladder-funding and first grants; inflection follows publication of rungs 1–3 results.

Sensitivity. Three variables dominate: (i) the rung-1 outcome, which multiplies or zeroes everything downstream — hence its position; (ii) whether frontier post-training pipelines adopt an intrinsic trigger at all (rows 1–2 fall to grant-scale without adoption, $\approx$ \$10–20M portfolio); (iii) thermodynamic hardware's own trajectory, entirely outside the lab's control (row 3 is an option, priced accordingly). Because the programme runs on the laboratory's own research team with AI-assisted execution, fully-loaded cost through 2028 is dominated by staff time and compute — making the asymmetry of central expected value to internal cost one to two orders of magnitude, before any post-2030 tail.

6 Risk Register and Falsification Discipline

Risk	Severity	Assessment and mitigation
Theoretical: clock convention unresolved	High	The corpus itself flags two candidate clocks (Fisher-Rao arc length vs Fisher action). Rung 1 must discriminate them; publish the resolution either way before scaling spend.
Theoretical: symmetry class of wild manifolds	High	ℓ is derived only once the symmetry class is fixed; for high-dimensional learned manifolds it is not established. Synthetic arenas dissolve this for rungs 1–3; treat any wild-system ℓ as provisional and say so.
Engineering: online arc-length estimation	Medium-High	The square root sits inside the integral; accumulated Fisher information is not a substitute. A cheap online estimator with a sub-resolution (ϵ) filter is the single most valuable engineering artefact; budget it explicitly (the corpus offers an ngc-learn implementation as first artefact).
Empirical: elasticity / just-so risk	Medium-High	The E. coli re-mapping episode shows the framework can survive a failed prediction by re-reading data as measurement — legitimate once, corrosive as habit. Mitigation: pre-registration of every rung, kill-criteria signed before data, publish falsifications with equal prominence.
Empirical: multiple-comparisons across domains	Medium	Eight domains invite pattern-hunting. Cap confirmatory claims to pre-registered tests; label all cross-domain readings (esp. §3.3) as heuristic in every output.
Competitive: 'good-enough' heuristics win	Medium	Agent-memory and continual-learning fields may standardise on simple schedules (self-distillation, idle-time consolidation) before principled triggers matter commercially. Mitigation: benchmark against those exact baselines; the pitch is matched-budget superiority or nothing.
Sociological: grand-theory scepticism; AI-assisted provenance	Medium	A single-author theory-of-everything with AI-drafted instalments will meet reflexive resistance. Mitigation: lead with rung results, not the vision; disclose drafting provenance proactively (the corpus already does); seek adversarial collaborators for rungs 6–7.
Reputational: societal overreach	Medium	Archetypal/metaphysical extensions in the synthesis documents are explicitly marked speculative; keep them quarantined from the empirical programme's branding.
Financial: valuation is conditional	—	All Table 2 figures are contingent on validation and adoption; the correct mental model is a portfolio of cheap options with pre-stated expiries, not a revenue plan.

Table 4 — Risk register. Severity is within-window impact on the programme thesis.

7 Recommended Actions for a Networked Frontier Laboratory

Months 0–3. Commission rung 1 (synthetic arenas) with pre-registered falsifiers; fund the online arc-length estimator as a deliverable in its own right; stand up the P1/P2 harness on an open agent-memory stack and a task-free class-incremental benchmark so rungs 2–3 start the day rung 1 reports. Simultaneously, socialise the instrument framing (one dial, three readings, stated falsifiers) with two or three trusted network nodes for adversarial review.

Months 3–9. Run rungs 2–4. Open the two partner conversations that gate the biological arm: the Levin laboratory / Allen Discovery Center ecosystem (planarian bistability intervals — rung 6) and an integrated mathematical-oncology group with adaptive-therapy trial data (rung 7 retrospective). Both conversations are materially easier with rung-1 results in hand. Approach the mortal-computation authors (Ororbis-Friston) and the Active Inference Institute — the corpus's own lineage — for the ngc-learn clock-module collaboration the machine-facing paper explicitly offers; and open a simulation-first channel to a thermodynamic-hardware maker via its public tooling (rung 5).

Months 9-18. Publish rungs 1-4 as a single strong-inference package regardless of direction; decision gate. On pass: fund rungs 5-8, begin standards work (an open Ω -measurement protocol is the network play), and initiate one licensing conversation in agent-memory. On fail: publish the falsification, retain the heuristic vocabulary where it is honestly useful, and close the empirical programme at minimal internal cost for a definitive answer — itself a creditable scientific product.

Standing posture throughout. Fund the ladder, not the vision; lead every external communication with a measurement, not a metaphor; keep the societal register explicitly heuristic; and hold the programme to the falsification discipline its own authors set — that discipline is, in the end, the most valuable and least imitable asset in the corpus.

Appendix A — Feasibility Rating Definitions

Rating	Definition (decisive validation within the 2026–2030 window)
A High	Kill-or-confirm achievable in ≤ 12 months by the in-house team with no gatekeeper dependencies; software-only or simulation-first.
B Medium	Decisive evidence in 1–3 years; requires one to two partners, specialised hardware, or wet-lab access.
C Low (in window)	Decisive evidence requires 3–5+ years, regulated or slow-cycle settings; only retrospective or preparatory work completes by 2030.
D Heuristic-only	Genuine interpretive value now, but rigorous validation is unlikely within the window; no confirmatory financial weight assigned.

Modifiers (+/–) indicate position within a band. Ratings assess validation feasibility, not commercial certainty.

Appendix B — Principal Sources

Corpus under assessment (github.com/alexsabine/Ryan_CRR): Sabine, *Phase-Gating Across Precision Channels* (AGI 2026, Springer LNCS 16855); *Mortal Computation Needs a Clock* (working paper v2, Aug 2026); *The Paradox That Must Not Be Resolved* (working paper, Aug 2026); *CRR: Canonical Formalism and Processual Extensions* (synthesis, 30 Aug 2026); *CRR as a Temporal Architecture for Thermodynamic and Mortal Computation* (synthesis, 30 Aug 2026).

Foundations referenced by the corpus: Hinton (2022) forward-forward / mortal computation; Ororbia & Friston (2023) *Mortal Computation*; Friston, Ramstead & Sakthivadivel (2026) Proc. R. Soc. A; Kleiner (2024); Kiefer (2025); Čencov (1982); Amari & Nagaoka (2000); Landauer (1961); Crooks (2007); Sivak & Crooks (2012); Winnicott (1971); Tronick et al. (1978); Mesman et al. (2009); Gopnik (2020); Takesian & Hensch (2013); Watson & Levin (2023); Tissot, Levin, Buckley & Watson (2024); Sandved-Smith et al. (2026).

2026 industry and research landscape (surveyed Aug 2026 for this dossier): Quanta Magazine, *Thermodynamic Computers Go With the (Energy) Flow* (Jul 2026); Extropic technical posts and US Dept. of Commerce CHIPS R&D letter of intent (Jul 2026); Whitelam, PRL 136 (2026); arXiv:2601.18699 and arXiv:2605.15053 on frontier-scale catastrophic forgetting; InfoWorld on self-distillation (Feb 2026); Mem0, *State of AI Agent Memory 2026* and *Dream*; Letta sleep-time compute (arXiv:2504.13171); coverage of Anthropic agent-memory consolidation and Google Memory Bank (May 2026); Omdia (Sep 2025), MarketsandMarkets, Grand View Research and Information Matters agentic-AI forecasts (2025–26); PatSnap and Adv. Intelligent Systems reviews of analog in-memory computing and neuromorphic drift (2025–26); Zhang et al. (2017) Nat. Commun.; West et al. survey of adaptive-therapy open questions (eLife 2023); Gallagher et al. (2024) Cancer Research; *Deriving Optimal Treatment Timing for Adaptive Therapy* (Bull. Math. Biol. 2025); Gatenby et al., Oncogene (2025); Levin laboratory publications incl. Pezzulo et al. (2021) on bistable somatic pattern memories and Levin (2019) on the computational boundary of a self; JMIR *Biocomputing: Beyond the Hype* (2026); fleet-scale robot learning (arXiv:2605.00416) and 2026 humanoid deployment analyses; Telnix/Hamming voice-AI latency benchmarks (2026); Shumailov et al. (2024) Nature, on model collapse.

Additional sources: Matthews, Ellsworth & Reasenber (2002) BSSA and USGS OFR 99-522 (BPT renewal model; generic aperiodicity $\alpha \approx 0.5$); HERP (2001) Japan $\alpha \approx 0.24$ and mainland-China fault studies $\alpha \approx 0.37$; Artemis Q2 2026 Catastrophe Bond & ILS Market Report (outstanding $\approx \$65.6\text{B}$; H1 issuance $\approx \$18\text{B}$; ILS limit $\approx \$141\text{B}$ per Aon Securities); Hamilton (1989) on regime switching; Scheffer et al. (2009) Nature, early-warning signals for critical transitions; predictive-maintenance market forecasts 2030 (Grand View, MarketsandMarkets, Research & Markets, IMARC: $\approx \$30\text{--}56\text{B}$; $>\$1\text{T/yr}$ unplanned-downtime problem).

Appendix C — The Mathematics in Plain Statistical Terms

This appendix is written for a first encounter: each piece of the mathematics restated in plain statistical language, so that four questions become reflexive for any finite system — what accumulates, what breaks, what does the rebuild draw on, and what do the dial Ω and the limit ℓ of that specific system say about its rhythm?

Object	What it says, statistically
Fisher-Rao distance	How many standard errors apart two states are — a z-score. Distance is measured in units of distinguishability, which is why it is unit-free and travels across domains.
Resolution $\epsilon = 1/\sqrt{N}$	A standard error: the smallest step the system can actually tell apart, given the evidence N it integrates per distinction. More evidence per judgement $\rightarrow$ finer grain.
Coherence C	A running odometer of distinguishable change: total arc length travelled so far, counted in standard-error units — accumulated effect size along the system's own path.
Rupture at $C \cdot \Omega = 1$	The regime ends after $\ell/\epsilon = 1/\Omega$ resolvable steps. Waiting for $1/\Omega$ quantal arrivals gives a Gamma interval with shape $1/\Omega$: $CV = \sqrt{\Omega}$, right-skew $2\sqrt{\Omega}$. Same counting logic as polls tightening with sample size — the more cells a regime holds, the steadier its rhythm.
Kernel $e^{C/\Omega}$	A softmax over the past with temperature Ω — Boltzmann weighting with coherence playing minus-energy. $\Omega \rightarrow 0$: argmax, one crystallised memory dominates every rebuild (habit). Ω large: near-uniform, the whole past votes (play). Saliency, not recency.
Measuring Ω	Read the dial off the rhythm: $\Omega = (CV \text{ of rupture intervals})^2$. Nothing is fitted — the denominator ℓ comes from symmetry class alone (π for two-state systems, 2π -scaled for cyclic ones), and the same Ω must govern trigger, kernel, and intervals or the grammar is falsified.
Scaffolding $\Omega_{\text{eff}} = \Omega/\sqrt{1-f}$	If a partner absorbs a fraction f of the distinctions a regime requires, your own count per regime shrinks by $(1-f)$; by counting statistics your effective temperature rises by $1/\sqrt{1-f}$. Shared work flattens the kernel — 'holding' as variance-pooling.
Landauer floor	Thermodynamics bills only irreversibility. Ideating futures from the archive is reversible — in principle, free. The cut that actualises one erases a bit: at least $k_B T \ln 2$. Fantasy is free; reality is paid in heat.

Table 5 — Each formal object of the grammar restated as a standard statistical idea.

Disclaimer. This dossier is an exploratory research-strategy analysis prepared from public materials. CRR is an unvalidated theoretical framework; nothing here asserts its correctness. All financial figures are illustrative, probability-weighted scenario ranges under stated assumptions, provided to support internal prioritisation only; they are not forecasts, valuations, solicitations, or investment, legal, or medical advice. Clinical readings in §3.1 are hypothesis-generating and are not treatment guidance.